R3.3 Electron sharing reactions

Radicals and how they are produced

Learning outcomes

By the end of this section you should be able to:

- (R3.3.1) Identify and represent radicals.
- (R3.3.2) Explain, including with equations, the homolytic fission of halogens.

Radicals and the ozone layer

Ozone, O_3, is a molecule concentrated in a layer above the Earth's surface. This ozone layer provides an important role in absorbing some of the ultraviolet (UV) radiation from the Sun, preventing it from reaching the Earth's surface.

You may have heard of a family of compounds called chlorofluorocarbons (CFCs) and how they can have a negative impact on our atmosphere. CFCs are easily broken down by UV radiation to form highly reactive radicals that react with the ozone causing the ozone layer to deplete. Regions where the ozone layer has been sufficiently depleted are known as 'holes'. These holes result in greater levels of UV radiation reaching the Earth's surface causing an increase in the diagnoses of skin cancers.

CFC compounds were commonly used as refrigerants, solvents and propellants in aerosol cans. Due to environmental concerns, many nations have banned the use of these compounds.

International Mindedness

When compounds used in industry and in the production of material goods have a negative impact on the environment, particularly the atmosphere, the effects are felt globally. The Montreal protocol (https://www.unep.org/ozonaction/who-we-are/about-montreal-protocol) was an international treaty developed in 1987 to reverse the effects of CFCs on the depletion of the ozone layer. This treaty has been considered to be one of the most successful international agreements. How might we use the Montreal protocol as an example to motivate other international agreements in the fight against climate change?

You can monitor daily ozone levels in our atmosphere using NASA's ozone watch (https://ozonewatch.gsfc.nasa.gov/).

It is clear we need to be well informed about radicals if they have the ability to negatively impact our health and our atmosphere. But what exactly is a radical? Why does a radical form in the presence of UV radiation? What species are able to form free radicals? Why are these species so highly reactive?

What are radicals?

Recall the Lewis formulas you drew in section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/). Some of these examples included diatomic halogens such as F_2, Cl_2, Br_2, I_2. Think about how the electrons were represented in Lewis formulas. A line represented a bond made up of two electrons and dots or x's represented the pair of non-bonding electrons. Electrons were either paired in a bond or part of a non-bonding pair. **Figure 1** shows the Lewis formulas for F_2, Cl_2, Br_2, I_2.

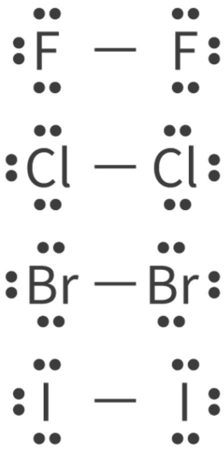

Figure 1. Lewis formulas of halogens.

More information for figure 1

The image displays Lewis structures of diatomic halogens, including fluorine (F2), chlorine (Cl2), bromine (Br2), and iodine (I2). Each molecule consists of a pair of identical halogen atoms joined by a single bond, represented by a line connecting the two atoms. Surrounding each atom are pairs of dots indicating non-bonding electrons, which align horizontally or vertically around the atomic symbol. For each diatomic molecule, these non-bonding electron pairs aim to fulfill the octet rule, using a combination of bonds and lone electron pairs. This depiction follows the conventional representation of electron sharing and pairing in the Lewis structure format.

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Can you recall an example where there was a single halogen atom by itself or a Lewis formula where there was a single, unpaired electron? Why do you think this is? There must be a reason why you are unlikely to encounter a species with a single, unpaired electron. Uncommon species tend to be ones that don't last long in the environment, indicating they are unstable and quick to react with other species.

A radical is a species that contains an unpaired electron. Radicals are always represented using a dot beside the species to indicate the unpaired electron. Some common radicals you might encounter are shown in **Figure 2**. Since electrons of opposite spin have the tendency to pair, an unpaired electron in a radical makes it highly reactive and unstable. A radical species rapidly forms a more stable covalent bond upon interaction with an electron from another substance.

$$\cdot Cl \qquad \cdot OH$$

$$\cdot O_2^{-} \qquad \cdot CH_3$$

Figure 2. Common radicals.

Radical species can be an atom, a molecule, a cation or an anion. **Figure 3** shows an example of each type of radical in comparison to a similar species that is not a radical.

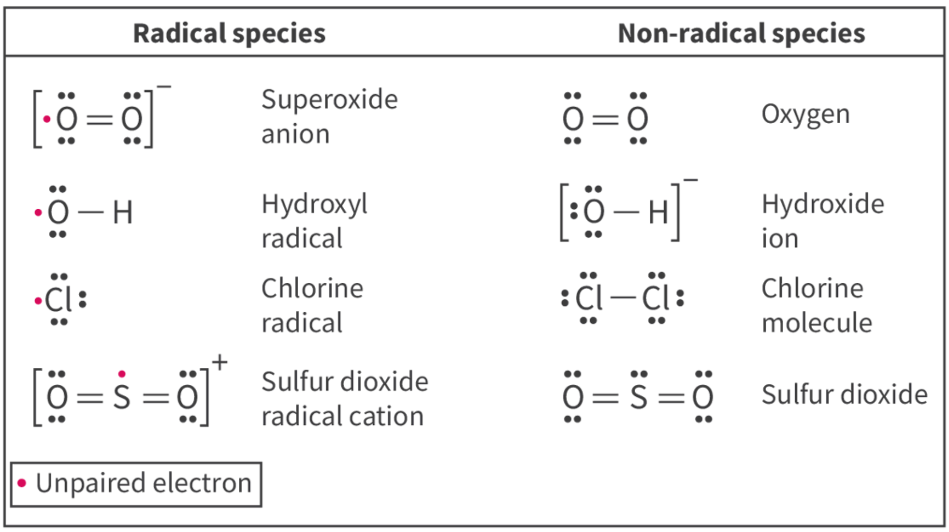

Figure 3. Lewis formulas of radical and non-radical species.

More information for figure 3

This diagram compares Lewis structures of radical and non-radical species. On the left side, under 'Radical species', it shows:

1. Superoxide anion: Represented as ([\cdot\displaystyle{\overset{\cdot}{\mathrm{O}}} = \displaystyle{\overset{\cdot}{\mathrm{O}}}]^{-}).
2. Hydroxyl radical: Illustrated as ([\cdot\displaystyle{\overset{\cdot}{\mathrm{O}}} - \mathrm{H}]).
3. Chlorine radical: Depicted as ([\cdot\displaystyle{\overset{\cdot}{\mathrm{Cl}}}]).

4. Sulfur dioxide radical cation: Shown as ([\cdot\displaystyle{\overset{\cdot}{\mathrm{O}}} = \displaystyle{\overset{\cdot}{\mathrm{S}}} = \displaystyle{\overset{\cdot}{\mathrm{O}}}]^{+}).

On the right side, under 'Non-radical species', it displays:

1. Oxygen: Shown as ([\displaystyle{\overset{\cdot}{\mathrm{O}}} = \displaystyle{\overset{\cdot}{\mathrm{O}}}]).
2. Hydroxide ion: Illustrated as ([\displaystyle{\overset{\cdot}{\mathrm{O}}} - \mathrm{H}]^{-}).
3. Chlorine molecule: Represented as ([\displaystyle{\overset{\cdot}{\mathrm{Cl}}} - \displaystyle{\overset{\cdot}{\mathrm{Cl}}}]).
4. Sulfur dioxide: Depicted as ([\displaystyle{\overset{\cdot}{\mathrm{O}}} = \displaystyle{\overset{\cdot}{\mathrm{S}}} = \displaystyle{\overset{\cdot}{\mathrm{O}}}]).

The diagram highlights that radical species have unpaired electrons, which are denoted by dots.

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Use **Interactive 1** to identify which species are radicals and which ones are not. Try counting the number of valence electrons and/or drawing the Lewis formula if you are finding the identification process difficult.

Radical	*Not* a radical

Cl_2 NO_2 OH^- NO OH CH_3 Cl CH_4 O_2^- O_2

Check

Interactive 1. Identification of radical species.

How radicals are formed

Recall from section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) that a covalent bond involves two atoms sharing two electrons. How might a radical be formed from species that contains covalent bonds? If enough energy is provided from heat or radiation, a covalent bond can break evenly with each atom having one of the electrons from the covalent bond. This type of bond breaking is called homolytic fission (**Table 1**).

In section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) you learned that UV radiation is the region of the electromagnetic spectrum that is just higher in energy than the visible spectrum. The energy provided by UV radiation is enough to cause the homolytic fission of some covalent bonds that have a relatively low bond enthalpy.

Table 1. Bond enthalpies for specific bonds that **do** and **do not** undergo homolytic fission fr radiation.

Bonds that undergo homolytic fission from UV radiation		Bonds that do not undergo homolytic fission from UV radiatio	
Bond	Bond enthalpy ($kJ\ mol^{-1}$)	Bond	Bond enthal ($kJ\ mol^{-1}$)
Cl—Cl	242	H—H	436
Br—Br	193	C—H	414
O—O	144	C—C	346

Look at section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet and predict what other bonds undergo homolytic fission in the presence of UV radiation?

Bonds that have a relatively low bond enthalpy can undergo homolytic fission in the presence of UV light. The most common are the covalent bonds between two oxygen atoms or between two halogen atoms. **Figure 4** shows the chemical equation representing the formation of a chlorine radical from chlorine in the presence of UV radiation.

$$Cl_2 \xrightarrow{UV} 2Cl\bullet$$

Figure 4. Lewis formulas of radical and non-radical species.

More information for figure 4

The image depicts a chemical equation illustrating the formation of chlorine radicals from chlorine molecules when exposed to UV light. On the left, "Cl_2" represents a chlorine molecule. An arrow labeled "UV" points from the chlorine molecule to the right, indicating the process of homolytic fission. On the right side of the arrow, "2Cl•" shows the formation of two chlorine radicals, represented by "Cl•", each with a single unpaired electron as indicated by the dot.

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Study skills

In chemistry, you might notice that sometimes there are things written on top of the arrow in a chemical reaction. Symbols written on top of the reaction indicate the conditions needed in order for the reaction to take place. For example, these conditions can include radiation, catalysts and solvents. You were introduced to these in section R2.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-of-reaction-id-43481/).

In this section UV is written on top of the arrow in the formation of a free radical as the process will not occur without this supplied energy.

Reaction mechanism for homolytic fission

In science and many other subject areas, the term mechanism is used to provide a detailed account of what is occurring. In chemistry a reaction mechanism is the detailed process of how substances are converted into other substances. You learned about these in section R2.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/) when analysing kinetic data. Mechanisms include a sequence of reactions for an overall chemical change and sometimes they include detail of how covalent bonds are broken and reformed using arrows to indicate electron movement.

The mechanism for homolytic fission uses a single-barbed arrow (sometimes called a fish-hook arrow) to show the movement of a single electron. In section R3.4.1–2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) you will look at reaction mechanisms involving a curly, two-barbed arrow, to show the movement of a pair of electrons. **Figure 5** shows the reaction mechanism for the formation of a chlorine radical. In this mechanism two single-barbed arrows are shown starting from the covalent bond moving to one of the chlorine atoms, creating two free radicals.

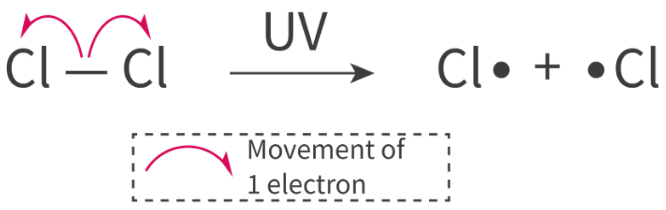

Figure 5. Reaction mechanism showing the formation of chlorine radicals.

More information for figure 5

This diagram illustrates the reaction mechanism for the formation of chlorine radicals. It shows a chlorine molecule (Cl-Cl) with two single-barbed arrows, indicating the movement of a single electron from the covalent bond towards each chlorine atom. Each chloride atom becomes a chlorine radical, noted as Cl•. There is a UV indication above the forward arrow between the reacting chlorine molecule and the resulting radicals. Below the main reaction, a dashed box explains the "Movement of 1 electron," with a visual representation of the single-barbed arrow to reinforce the concept of electron movement in homolytic fission.

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Theory of Knowledge

Over time, overexposure to the UV rays from the Sun causes damage to the skin including sunburn, changes to pigmentation and even skin cancer. One way to reduce sun exposure is by using sunscreen. Some sunscreens work by using chemicals that are capable of absorbing UV light, preventing the UV rays from reaching the skin. There are some effective sunscreen ingredients that are known to have harmful effects on human health and aquatic ecosystems because they produce free radicals in their absorption of UV rays. Does knowledge of these harmful ingredients compel the user to seek a responsible, alternative option?

UV rays cause the degradation of CFCs creating free radicals in the atmosphere. CFCs are a category of organic compounds that contain carbon, chlorine and fluorine atoms covalently bonded. The free radical formation typically occurs when UV light causes homolytic fission in the carbon–chlorine bonds, but occasionally homolytic fission can break the carbon–fluorine bond. Examine **Table 2** that compares some properties of the C–F and C–Cl bonds.

Table 2. Properties of the C—F and C—Cl bonds.

Bond	Bond length (10^{-12} m)	Bond enthalpy (kJ mol^{-1})
C—F	138	492
C—Cl	177	324

Bond length is the distance between the two nuclei that make up the covalent bond and bond enthalpy is the amount of energy required to break the bond and bond. Notice the relationship between the length of the bond and the amount of energy required to break it. The C–F bond is shorter and stronger, therefore it does not break as easily or frequently as the C–Cl bond. Therefore, in CFCs, the free radicals produced are typically Cl• free radicals and less frequently F• free radicals.

When the chlorine free radicals are formed by the breakdown of CFCs, they are then capable of reacting with ozone molecules, O_3, in the atmosphere. Ozone molecules contain three oxygen atoms that are held together by covalent bonds. Depleting the protective ozone layer occurs by the reaction sequence in **Figure 6**.

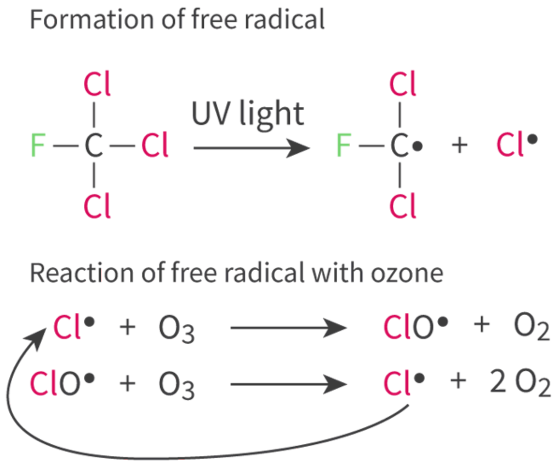

Figure 6. Depletion of ozone by CFCs.

More information for figure 6

The diagram illustrates two main processes: the formation of free radicals and their reaction with ozone.

In the formation of free radicals, the diagram shows that CFC (Chlorofluorocarbon) molecules, represented with a chemical structure, are broken down by UV light. The structure includes chlorine (Cl), carbon (C), and fluorine (F) atoms. Initially, the CFC molecule (F-C-Cl, with additional Cl bonds) is exposed to UV light. This causes the bond between the chlorine and carbon to break, resulting in the formation of a chlorine free radical (Cl•) and another chemical fragment (F-C• with Cl).

In the reaction with ozone, the diagram shows how chlorine free radicals (Cl•) interact with ozone molecules (O3) in the atmosphere. The chlorine radicals react with ozone to form chlorine monoxide (ClO•) and molecular oxygen (O2). This sequence continues; the ClO• can further react with another O3 molecule to regenerate Cl• and produce two O2 molecules. This cycle represents how the presence of chlorine free radicals can lead to continuous depletion of ozone in the atmosphere.

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The atmosphere also contains the allotrope of oxygen that you are probably more familiar with, O_2. Free radicals formed in the decomposition of CFCs, however, do not react with O_2. Why do you think this is? Examine the bond enthalpy values in **Table 3**.

Table 3. Bond enthalpy values.

Type of bond	Bond enthalpy (kJ mol^{-1})
O—O single bond	144
O=O double bond	498

Type of bond	Bond enthalpy ($kJ\ mol^{-1}$)
Bonds in O_3	363

Recall from section S2.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) that double bonds are stronger than single bonds, therefore requiring more energy to break the bond as shown in the table. The bonds for ozone, however, are in between the value of a single and double bond, indicating that ozone requires less energy to break the bond than O_2. Therefore, free radicals are able to break the 1.5 bonds in ozone, but are unable to break the double bond in an O_2 molecule.

In reactions that involve radicals homolytic fission is required before the reaction can proceed. This is known as initiation and involves a covalent bond undergoing homolytic fission to form radicals. This process always requires energy and is therefore endothermic.

Worked example 1

Sketch the reaction mechanism for the formation of a radical from bromine. State any conditions required.

In the reaction mechanism for the formation of a radical, a single-barbed arrow is needed to show the movement of a single electron.

To form a bromine radical, bromine would need to undergo homolytic fission in the presence of UV radiation or heat.

In a reaction mechanism involving radicals remember to include single-barbed arrows to show the movement of a single electron. In the formation of a radical the arrow should start at the covalent bond and should end at the atom that leaves with the single electron to form a radical. Do not use a curly arrow (double-barbed arrow) in a reaction mechanism involving radicals.

You have just learned how homolytic fission makes free radicals, what do you think could happen when two free radicals meet? Recall from section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) when you learned how a covalent bond is formed by the overlapping of atomic orbitals containing an unpaired electron. The reverse process of homolytic fission is the formation of a covalent bond. You will see the pairing of free radicals in the next section.

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:** Thinking skills — Engaging with, and designing, linking questions
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Group activity (3—4 students)

Linking questions are used throughout the DP chemistry syllabus to highlight links between course concepts with the intention of looking at a concept with a different perspective.

From the list, select one of the previously learned concepts. Each group member should select a different understanding. Your task is to design a linking question between the concepts and understandings from this section about radicals and your chosen syllabus understanding.

Previous concepts:

- Lewis formulas (see section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/)).
- Electromagnetic spectrum (see section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/))
- Bond enthalpy (see section R1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-breaking-and-forming-id-43473/)).
- Energy transfer between the system and the surroundings (see section R1.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/energy-transfer-in-chemical-reactions-id-43469/)).
- Select a different concept from previously learned material.

After each group member has developed a linking question, share and discuss each of these questions as a group.